

Learning on $\mathbb{B}^n$ and Equivariant Architectures

This chapter studies learning problems on the Boolean hypercube $\mathbb{B}^n$ through the lens of symmetry, spectral structure, and architecture design. The central theme is that many target functions, hypothesis classes, and training objectives become substantially more tractable when one exploits invariances or equivariances under a group action rather than treating inputs as unstructured bit strings.

A useful starting point is the notion of symmetry-aware generalization error. For a hypothesis class constrained by a symmetry group G , one may write the error profile as $\varepsilon(r)$, where r indexes the resolution, radius, or complexity scale at which the learner is evaluated. In this setting, $\varepsilon(r)$ is not merely a scalar performance summary; it records how well the model class captures the orbit structure induced by G , and how rapidly approximation improves as the model is allowed to represent finer symmetry-respecting features. This viewpoint aligns naturally with the book's broader emphasis on closure: the relevant question is not only whether a model fits data, but whether the class of admissible predictors is closed under the transformations that the task itself regards as irrelevant.

Fourier analysis on $\mathbb{B}^n$ provides a canonical language for making these ideas precise. Boolean functions admit expansions in the Walsh–Fourier basis, and symmetry constraints often manifest as sparsity or concentration in particular spectral coordinates. In favorable cases, the target function behaves like a junta, depending effectively on only a small subset of coordinates, or more generally exhibits sparse Fourier support. Such structure can be exploited both for learning and for compression: a learner that identifies the relevant coordinates or dominant coefficients can achieve strong sample efficiency and interpretability. From the perspective of this book, Fourier sparsity is a closure phenomenon in disguise, since the admissible hypothesis class is effectively reduced to a stable subfamily determined by the symmetry and support constraints.

The spectral viewpoint also clarifies why symmetry-aware learning can be robust to overparameterization. When the target lies near a low-degree or low-support Fourier subspace, the effective dimension of the problem is much smaller than 2^n , and the learner can focus on the orbit representatives or coefficient patterns that matter. In practice, this means that the learning pipeline may first identify a symmetry class, then estimate the corresponding spectral mass, and only afterward instantiate a predictor at the desired resolution. The chapter's organizing metric $\varepsilon(r)$ is therefore best read as a multi-resolution error curve: as r increases, the learner is permitted to resolve finer Fourier structure, and the error should decrease in a way that reflects the underlying symmetry class.

Group-equivariant networks provide a constructive architecture for encoding these constraints directly into the model. For permutation symmetries, equivariant layers ensure that permuting the input induces a corresponding permutation of intermediate representations and outputs. More generally, wreath products and related semidirect constructions describe layered symmetry actions that arise when local symmetries are combined with global rearrangements. In such architectures, each layer is designed so that the symmetry constraints are preserved across composition, yielding a network whose entire forward pass respects the chosen group action. This closure across layers is essential: if equivariance is enforced only at the input or output, the internal representation may drift outside the intended symmetry class, weakening both identifiability and generalization.

The closure of constraints across layers also raises identifiability questions. When multiple parameter settings realize the same equivariant map, one must distinguish genuine degrees of freedom from redundant parametrizations induced by symmetry. This is especially important in model families with tied weights, orbit pooling, or structured tensor factorizations. A well-designed equivariant architecture should make the symmetry constraints explicit enough that the learned representation

is identifiable up to the unavoidable action of the symmetry group. In practice, this means that the architecture should separate task-relevant variation from symmetry-induced equivalence classes, so that optimization and interpretation remain stable under the group action.

A useful design principle is to treat equivariance as a layerwise invariant rather than a global afterthought. If each layer preserves the relevant group action, then the composition of layers remains inside the same admissible family, and the model can be reasoned about as a closed system. This is particularly important for permutation-equivariant networks and architectures built from wreath products, where the symmetry may act on nested or hierarchical index sets. In those settings, closure across layers is not just aesthetically pleasing; it is what prevents the network from silently encoding spurious asymmetries that would otherwise undermine the intended inductive bias.

The chapter also points toward an artifact-driven workflow. An equivariant model zoo can serve as a catalog of architectures indexed by symmetry type, input domain, and spectral bias. Such a repository is useful not only for benchmarking but also for distillation: a large equivariant model may be compressed into a smaller circuit-like representation that preserves the relevant symmetry behavior while reducing computational cost. This distillation perspective connects learning back to the book’s circuit and closure themes, since the resulting artifact can be analyzed as a finite compositional object with explicit invariants. In a mature workflow, the model zoo is not merely a collection of examples; it is a reproducible design space in which symmetry assumptions, parameter sharing schemes, and compression targets can be compared systematically.

Distillation to circuits is especially valuable when the goal is deployment or auditability. A circuit-style artifact makes the learned symmetry constraints explicit, enabling inspection of which features are preserved, which are discarded, and how the approximation error is distributed across scales. This is consistent with the chapter’s emphasis on $\varepsilon(r)$: the distilled artifact should come with a resolution-dependent error profile, not just a single aggregate score. Such a profile helps distinguish coarse but stable approximations from fine-grained models that may be accurate only in a narrow regime.

Privacy considerations enter naturally once symmetry-aware learning is deployed on sensitive data. Symmetry can reduce the effective degrees of freedom of a model, but it can also concentrate information in structured parameters or shared representations. Consequently, one must ask whether equivariant inductive bias improves or weakens privacy guarantees in a given setting. A careful treatment should examine how orbit structure, parameter sharing, and distillation interact with leakage, memorization, and membership inference. The practical lesson is that symmetry is not automatically privacy-preserving; rather, it changes the geometry of the learning problem in ways that must be audited alongside accuracy.

From a systems perspective, privacy analysis should be carried out together with the architecture design. If a model zoo is used to select among symmetry-respecting architectures, then the selection criterion should include not only test accuracy and compression ratio, but also the extent to which the architecture exposes sensitive correlations through shared weights or low-dimensional latent factors. Likewise, if a model is distilled into a circuit-like artifact, the distillation step may either reduce or amplify leakage depending on how faithfully it preserves memorized structure. The right conclusion is not that equivariance is unsafe, but that its privacy impact is architecture-dependent and must be measured explicitly.

Taken together, the chapter develops a unified picture: Fourier structure explains why some Boolean learning problems are compressible; equivariant architectures show how to encode the relevant invariances by design; closure across layers ensures that symmetry constraints survive composition; and artifact-oriented distillation makes the resulting models inspectable and reproducible. The privacy dimension reminds us that every structural gain comes with a corresponding systems-

level responsibility. In the language of this book, learning on $\mathbb{B}^n$ is most effective when symmetry, composition, and closure are treated as first-class design principles rather than as after-the-fact regularizers.